

MATH 1110-College Algebra

The following sets of three statements contain two truths and a lie. Your task is to determine which are truthful statements and which is the lie. Provide justification for your conjectures. (Letters represent positive real numbers.)

Radicals

A) $\sqrt{x^2 + y^2} = \sqrt{x^2} + \sqrt{y^2} = x + y$

B) $\sqrt{\frac{36}{x^2}} = \frac{\sqrt{36}}{\sqrt{x^2}} = \frac{6}{x}$

C) $\sqrt{a^2 b^2 c^4 d^8} = abc^2 d^4$

Simplification

A) $\frac{x}{x(x-4)} = \frac{1}{x-4}$

B) $\frac{y}{a+b} = \frac{y}{a} + \frac{y}{b}$

C) $\frac{x+y}{b} = \frac{x}{b} + \frac{y}{b}$

Exponents

A) $x^5 x^2 = x^7$

B) $(a^2)^3 = a^6$

C) $\frac{x^6 y^4}{x^3 y} = x^2 y^4$

Expanding Binomials

A) $(x+2y)^2 = x^2 + 4y^2$

B) $(2a+3b)^2 = 4a^2 + 12ab + 9b^2$

C) $(-4x+1)^2 = 16x^2 - 8x + 1$

Sidework

In order to be successful in Section 1.3 your factoring skills need to be sharp. Work on factoring the following expressions in order to improve your skills.

1. $4x + 8y + 16z$

2. $3xy^2 + 6x^3y - 15x^2$

3. $\frac{x^2 + x}{x}$

4. $x^2 - 4$

5. $x^2 + 5x + 6$

6. $x^2 - 2x - 8$

7. $2x^2 - 5x - 3$

8. $\frac{x^2 - 16}{2x^2 + 7x - 4}$